Research Proposal

Time-Changed Lévy Models for Option Pricing and Implied Volatility Surfaces

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MATH70088 – Statistics Research Project

2025–26

1. Introduction

A standard result from Black–Scholes is that volatility should be constant, but this is not what we see in real option markets. Implied volatility changes across both strike and maturity, creating a non-flat implied volatility surface. In practice, this means that simple diffusion models do not explain the skew or smile in option prices very well, especially when returns are asymmetric or heavy-tailed.

Time-changed Lévy models are a natural way to deal with this problem. They combine a Lévy process, which can capture jumps and non-Gaussian returns, with a stochastic time change, which can represent changing market activity or volatility clustering (Carr and Wu, 2004; Carr et al., 2003). This makes them a good framework for studying option prices across the whole IV surface.

In this project, I will compare two time-changed Lévy models with their pure Lévy versions:

- NIG + Gamma subordinator,
- CGMY + Gamma subordinator,
- pure NIG and pure CGMY as baselines.

The main questions are:

1. Does adding a time change improve fit to the IV surface and next-day pricing performance?
2. How do NIG and CGMY differ across different parts of the surface, such as short and long maturities or out-of-the-money options?
3. Are any improvements statistically significant?

The project does not try to build a completely new model. Instead, its contribution is to give a careful comparison of a small number of relevant models under one consistent framework, using both in-sample fit and out-of-sample testing.

2. Literature review

The main starting point is Carr and Wu (2004), who develop a general option pricing framework based on time-changed Lévy processes. In their setup, the Lévy driver captures jumps, while the stochastic time change captures stochastic volatility. This makes their paper directly relevant to the present project.

A related paper is Carr et al. (2003), which studies stochastic volatility for Lévy processes and shows why it is useful to combine jump behaviour with time-varying volatility. This is one reason why comparing pure Lévy models with time-changed versions is interesting.

For the choice of drivers, I focus on NIG and CGMY. The NIG model of Barndorff-Nielsen (1997) is a well-known skewed heavy-tailed model from the generalized hyperbolic family. It provides a relatively simple but flexible benchmark. The CGMY model of Carr et al. (2002) is more flexible in its jump structure and is widely used in Lévy-based option pricing.

Another practical reason for choosing these models is that both have closed-form characteristic functions. This makes Fourier pricing possible, which is important because calibration requires repeated pricing of many options. For computation, I will use the FFT approach of Carr and Madan (1999). I also use Huang and Wu (2004) as motivation for treating model specification itself as an empirical question rather than assuming one model is automatically best.

Compared with the earlier literature, this project puts more weight on a clear model-versus-baseline comparison and on out-of-sample performance, rather than only in-sample fit.

3. Methodology

3.1 Model setup

The asset price will be modelled under the risk-neutral measure as

$$\log S_t = \log S_0 + (r-q)t + Y_t, \qquad Y_t = X_{\tau(t)} + \Delta_{\mathbb{Q}}(t),$$

where X is the Lévy driver, $\tau(t)$ is an independent stochastic clock, and $\Delta_{\mathbb{Q}}(t)$ is a drift adjustment chosen so that the discounted price process is a martingale under $\mathbb{Q}$ (Carr and Wu, 2004).

I will focus on *independent* time change. This is mainly for tractability: it keeps the characteristic function manageable and makes calibration realistic within the available time.

3.2 Pricing

The main analytical object will be the characteristic function of the log-price process. European option prices will then be computed using Fourier methods. The main numerical method will be the Carr–Madan FFT (Carr and Madan, 1999). I will also use the COS method as a check on selected strike–maturity points to make sure the numerical results are stable.

3.3 Calibration

On each trading day, model parameters will be calibrated to option data using a vega-weighted IV loss:

$$\min_{\theta} \sum_{(k,T)} w_{k,T}\left(\mathrm{IV}^{model}_{k,T}(\theta) - \mathrm{IV}^{mkt}_{k,T}\right)^2, \qquad w_{k,T} \propto \mathrm{Vega}_{k,T}.$$

I will also use a vega-normalised price-based loss as a robustness check. The IV-based loss is useful because the implied volatility surface is the main object of interest, while the price-based loss helps check that the results are not driven by Black–Scholes inversion alone.

Calibration will be done with constrained optimisation and multiple starting values. This is important because models such as CGMY can be sensitive to starting values and may have local minima.

3.4 Evaluation

The model comparison will use three types of evaluation:

- **Surface comparison:** market IV surface, model IV surface, and the error surface $\Delta IV = IV^{model} - IV^{mkt}$.
- **Out-of-sample testing:** calibrate on day t and evaluate on day $t+1$ using a daily scalar loss such as vega-weighted IV RMSE.
- **Statistical testing:** compare daily loss differences using the Diebold–Mariano test with HAC/Newey–West variance (Diebold and Mariano, 1995; Newey and West, 1987). Within a single day, I will also use maturity-block bootstrap confidence intervals for the loss difference.

This setup should make it possible to separate in-sample fit from actual predictive performance.

4. Feasibility and timeline

The project is feasible because I already have access to multi-year option chain data, underlying prices, and the market information needed to construct moneyness, maturities, and implied volatilities. The scope is also limited on purpose: only two main time-changed models and two baselines are included.

The planned timeline is:

May	Clean the data, apply liquidity filters, build market IV surfaces, and implement pure NIG and pure CGMY baselines.
June	Implement pricing for NIG+Gamma and CGMY+Gamma, and carry out numerical stability checks.
July	Run calibration across many days, produce model and error surfaces, and check convergence and identifiability.
August	Carry out out-of-sample testing, run statistical tests, and complete the final write-up.

The main risks are unstable calibration, weak parameter identification, and numerical error. I will try to reduce these risks using parameter constraints, multiple starting values, and cross-checks between FFT and COS pricing. If time is limited, I will keep the main comparison and treat extensions such as NIG with an IG subordinator as future work.

5. Limitations and significance

The main limitation is that I only consider independent time change, so the model does not include leverage-type dependence between returns and volatility activity. This is a deliberate choice to keep the project manageable.

The main value of the project is that it gives a careful empirical comparison of whether time change improves option pricing in practice, rather than only in theory. Even if the result is that the gain is limited, that is still useful, because it helps show when added model complexity is or is not justified.

There are no major ethical issues, since the project uses financial market data only. The main requirement is accurate citation of previous work and honest reporting of limitations.

References

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